

Topic1 Recap

- Current is the rate of flow of electrons in a conductor.

$$i = \frac{dq}{dt} \text{ (A)}$$

$$q = \int_{t_0}^{t_1} i \, dt + q(t_0), \quad t_0 \leq t \leq t_1 \text{ (C)}$$

- Voltage is the potential difference between two points of the circuit (difference in charge between two locations):

$$v_{ab} = v_a - v_b = \frac{dw}{dq} \text{ (V)}$$

- Current/voltage are described by their values and direction/polarity.
- Power and energy

$$p = \frac{dw}{dt} = vi \text{ (W)}$$

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt, \quad t_0 \leq t \leq t_1 \text{ (J)}$$

Topic1 Recap

- Passive sign convention
 - Power is **positive** if the current enters the **positive terminal**, $p = +vi$.
Power is **negative** if the current enters the **negative terminal**, $p = -vi$.
 - Positive power is **absorbed/dissipated** by an element.
Negative power is **supplied/generated** by an element.
- Conservation of energy
 - $\sum p = 0$
- Sources
 - Voltage sources generate or dissipate power at a specified voltage with whatever current is required.
 - Current sources generate or dissipate power at a specified current with whatever voltage is required.
 - Sources can be ideal/real, dependent/independent.

Topic 1 Recap

Resistors

- Resistance R is a physical property of a circuit element that impedes the flow of charge. It is measured in ohms, Ω .
- Ohm's Law establishes the relationship between the voltage across a resistive element (called resistor) and the current through it:

$$v = Ri$$

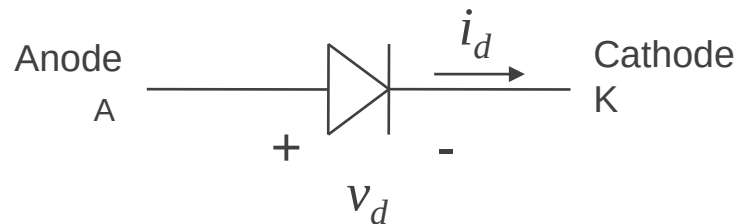
- Short circuit: $R = 0 \Omega \rightarrow v = Ri = 0$. Open circuit: $R = \infty \Omega \rightarrow i = \frac{v}{R} = 0$.
- Conductance $G = \frac{1}{R}$
- Power dissipation on resistor:

$$p = Ri^2 = \frac{v^2}{R}$$

Topic 1 Recap

Diodes

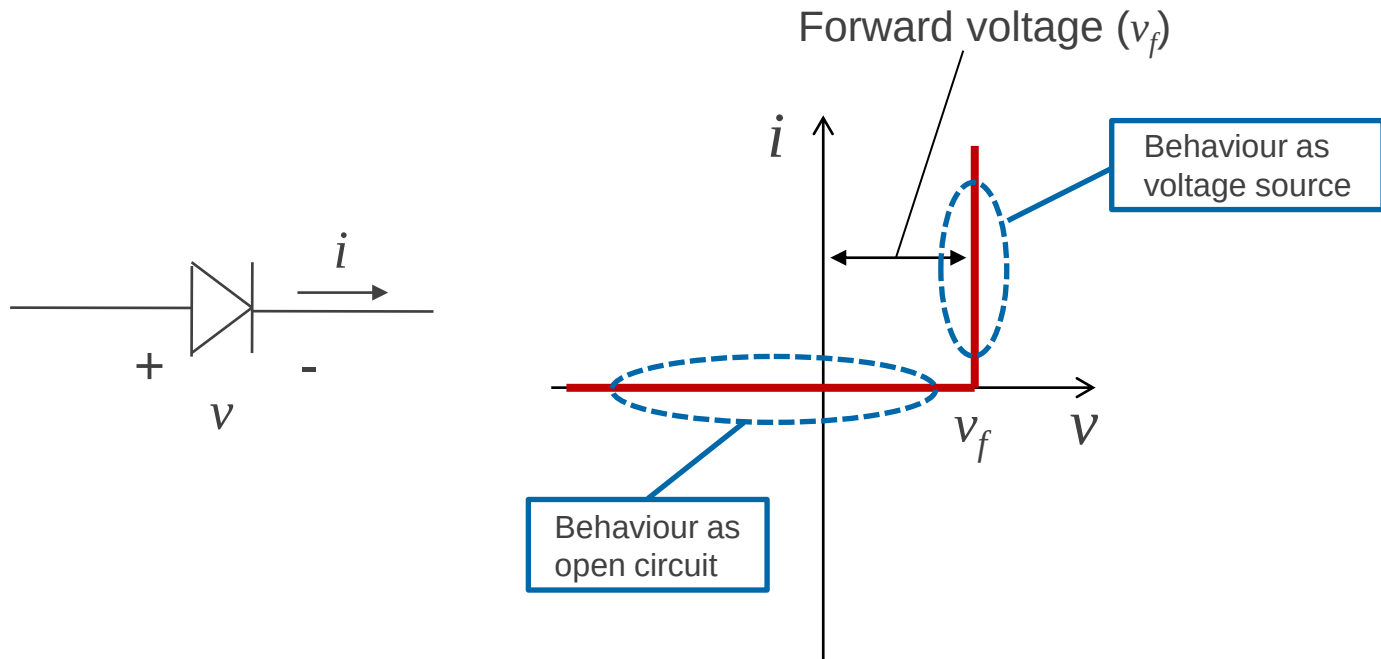
- A diode is a semiconductor electrical component that allows the flow of current in only one direction.
- Current flows in the direction of the arrow, and not against the direction of the arrow.
- Voltage is required to start current flow in the forward direction.
 - When $v_d >$ forward voltage, the diode is **forward biased**, and $i_d > 0$.
 - When $v_d <$ forward voltage, the diode is **reverse biased**, and $i_d \approx 0$.



Topic 1 Recap

Diodes

- Diodes have a *non-linear* v - i characteristic.
 - When the diode is **forward biased**, it behaves as a **voltage source of value v_f V**.
 - When the diode is **reverse biased**, it behaves as an **open circuit ($i = 0$)**.



Useful information: Light Emitting Diodes (LEDs)

- A Light Emitting Diode (LED) is a semiconductor electrical component that emits light when a forward current is passed through it.
- Forward current refers to the flow of current from the anode (positive) to the cathode (negative) terminals.
- Direction of the current is specified because electrical polarity of the LED is important, and will only emit light in forward-bias. LEDs will not illuminate in reverse-bias.

